Suggested problems: 1-17 odd, 23-29 odd, 35, 37, 41, 43

Problem 1: The temperature, in °C, at a point (x, y, z) is given by $T(x, y, z) = x^2 + y^2 + z^2$. What do the level surfaces of T look like and what do they mean in terms of temperature?

Problem 2: What do the level surfaces of the following functions look like?

(a) $f(x, y, z) = x^2 + y^2$

(b) $g(x, y, z) = z - y$

Problem 3: A company makes three sizes of cardboard boxes: small, medium, and large. It costs \$2.50 to make a small box, \$4.00 for a medium box, and \$4.50 for a large box. The fixed costs to make any number of boxes is \$8000.

(a) Express the cost, C, of making x small boxes, y medium boxes, and z large boxes as a function $C = f(x, y, z)$.

(b) Find $f(3000, 5000, 4000)$ and interpret it.

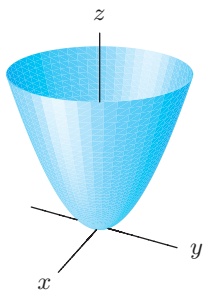

Figure 12.72: Elliptical paraboloid $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

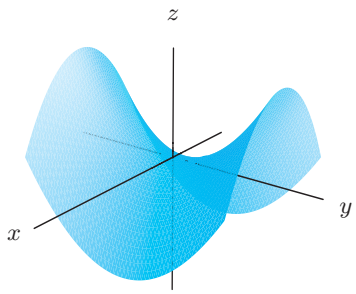

Figure 12.73: Hyperbolic paraboloid $z = -\frac{x^2}{a^2} + \frac{y^2}{b^2}$

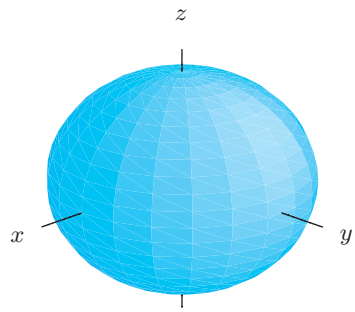

Figure 12.74: Ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

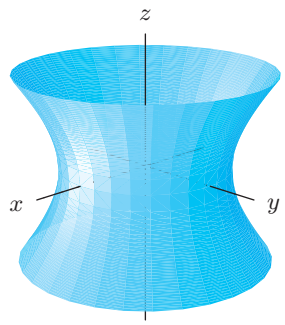

Figure 12.75: Hyperboloid of one sheet $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

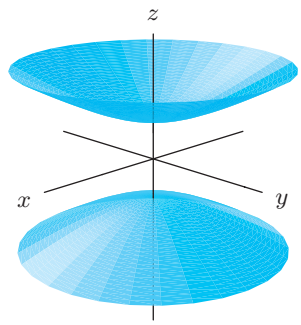

Figure 12.76: Hyperboloid of two sheets $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$

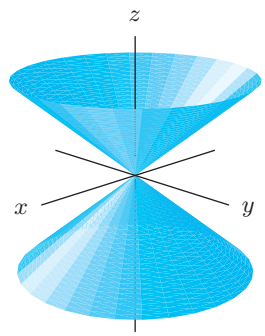

Figure 12.77: Cone $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$

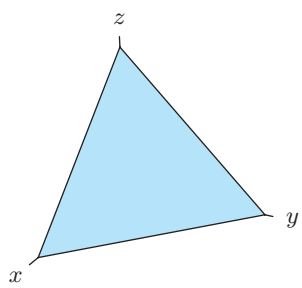

Figure 12.78: Plane $ax + by + cz = d$

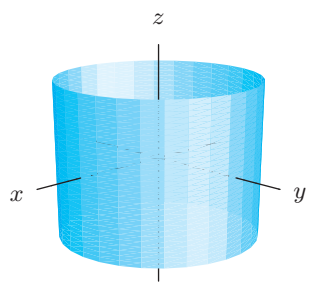

Figure 12.79: Cylindrical surface $x^2 + y^2 = a^2$

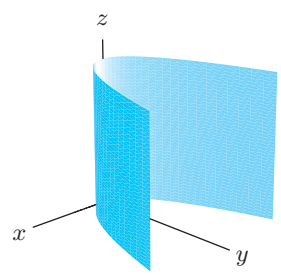

Figure 12.80: Parabolic cylinder $y = ax^2$

Problem 4: Use the figures above to identify the surfaces defined by the following equations.

(a) $z - 9x^2 - y^2 = 0$

(b) $z + x^2 - y^2 = 0$

(c) $-x^2 - 4y^2 + 1 = 0$

(d) $y - 2x^2 = 0$

Problem 5: Describe the level surfaces of

$$f(x, y, z) = \frac{x^2}{4} + z^2$$

in words.

Problem 6: Let

$$f(x, y, z) = x^2 - y^2 + z^2.$$

Describe the level surface $f(x, y, z) = 1$ in words.

Problem 7: Find a function $g(x, y, z)$ whose level surface with $g = 0$ is the graph of $f(x, y) = x + 2y$. Now find a different function $h(x, y, z)$ whose level surface with $h = 1$ is the graph of $f(x, y) = x + 2y$.

Problem 8: Define $g(x, y, z) = x^2 + y^2 + z^2$. Can the level surface defined by $g = 10$ be written as the graph of a single function $f(x, y)$?